Enhancing Digital Media Integrity: A Natural Language Processing Model for Clickbait Detection and Mitigation

Carrie Li Mark Yan Minyu Deng Zhiyuan Li
University of Pennsylvania

Abstract

This paper investigates automatic clickbait detection using modern natural language processing techniques, with an emphasis on efficiency and knowledge transfer. We evaluate a range of models, from simple logistic regression baselines to fine-tuned transformer-based architectures. Our primary contribution is the application of knowledge distillation from a large language model (GPT-4o-mini) to a compact DistilBERT student model, using both LLM-generated and human-annotated soft labels. Experimental results on the Webis Clickbait Corpus 2017 show that distillation substantially improves DistilBERT performance, allowing it to match or slightly surpass a BERT-base baseline in accuracy while using significantly fewer parameters. These findings demonstrate the effectiveness of LLM-guided distillation for practical clickbait detection.

1 Introduction

In today's digital media landscape, clickbait has emerged as a pervasive phenomenon that undermines the quality and credibility of online content. Clickbait refers to content that deliberately employs sensational, misleading, or incomplete language to exploit readers' curiosity and entice them to click on links [6]. This practice not only degrades user experience but also serves as a vehicle for misinformation, eroding public trust in media institutions.

From a natural language processing (NLP) perspective, clickbait detection presents a compelling challenge that extends beyond simple keyword identification. Effective detection requires understanding deep semantic structures, emotional undertones, and rhetorical devices such as exaggeration, suspense, and curiosity gaps. Furthermore, as language evolves dynamically with culture and context, clickbait expressions continuously adapt, making automatic identification an ongoing challenge.

Non-Clickbait:
"Senate Passes Infrastructure Bill with $1.2 Trillion in Funding"
→ *Factual, provides specific details.*
Clickbait:
"You Won't Believe What Congress Just Did!"
→ *Creates curiosity gap, withholds content.*

Figure 1: Comparison of non-clickbait and clickbait headlines. Clickbait uses rhetorical strategies to provoke curiosity while withholding key information.

Figure 1 illustrates the distinction between clickbait and non-clickbait headlines. The key difference lies in information presentation: non-clickbait headlines provide complete, factual information, while clickbait headlines deliberately withhold details to create a "curiosity gap."

We formalize clickbait detection as a binary classification task. Given a headline h, the goal is to predict $y \in \{0, 1\}$, where $y = 1$ indicates clickbait:

$$f : \mathcal{H} \to \{0, 1\} \tag{1}$$

where $\mathcal{H}$ denotes the space of all possible headlines.

In our extended framework, we additionally leverage soft labels from human annotators ($s_{\text{human}} \in [0, 1]$) and LLM predictions ($s_{\text{LLM}} \in [0, 1]$) to provide richer supervision signals during training through knowledge distillation.

We selected this task for several reasons. First, it addresses a real-world problem with tangible social impact. Second, the task provides an ideal testbed for knowledge distillation: clickbait detection requires nuanced linguistic understanding that LLMs possess, yet practical deployment demands smaller, efficient models. Third, the Webis Clickbait Corpus 2017 includes both binary labels and continuous human judgment scores, enabling multi-source supervision experiments. Finally, the

inherent class imbalance (3:1 ratio) presents realistic challenges common in NLP applications.

2 Literature Review

Clickbait detection has evolved from rule-based heuristics to sophisticated neural approaches. This section reviews foundational work and recent advances.

Foundational Work. Potthast[6] introduced the first publicly available corpus and computational model for automatic clickbait detection. They compiled a Twitter dataset of 2,992 tweets from major U.S. publishers, annotated by three assessors, with 767 samples labeled as clickbait (Fleiss' $\kappa = 0.35$). Based on analyses of the "curiosity gap," the authors constructed 215 lexical, stylistic, syntactic, and pragmatic features, training a Random Forest classifier that achieved ROC-AUC of 0.79 and precision/recall around 0.76. This work established clickbait detection as a supervised learning task and provided the foundational dataset that evolved into the Webis Clickbait Corpus 2017 used in our project.

Pre-trained Language Models. Han[2] examined large pre-trained transformers—BERT, RoBERTa, and XLNet—for clickbait detection, framing it as binary classification or regression. Their survey covers domain-adaptive pretraining, language-specific models (e.g., IndoBERT, BanglaClickBERT), and hybrid pipelines integrating PLM encoders with auxiliary modules. Fine-tuned PLMs achieve F1/AUC typically in the 0.8–0.9 range, surpassing traditional SVMs and Random Forests. However, challenges remain: sensitivity to domain shift, limited interpretability, and reliance on large annotated corpora. This motivates our choice of DistilBERT as the student model.

LLM-based Approaches. Lihi[4] present an interpretable two-stage framework integrating detection and tactic attribution. They construct synthetic tactic-annotated headlines using GPT-4o and compare fine-tuned BERT with LLMs under zero-shot and few-shot prompting. Results show fine-tuned BERT significantly outperforms prompt-based LLMs (F1 = 0.91 vs. $\leq$ 0.47). This aligns with our observation that gpt-4o-mini achieved only F1 = 0.44 in zero-shot classification. However, we demonstrate that LLM knowledge can be effectively transferred through distillation.

Knowledge Distillation. Hinton[3] introduced knowledge distillation where a smaller student model learns from a larger teacher's soft probability distributions, which encode richer information than hard labels. Sanh[7] applied this to create DistilBERT, retaining 97% of BERT's performance while being 60% smaller and faster. Our work extends this paradigm by using gpt-4o-mini as teacher and combining its soft labels with human annotation distributions.

3 Experimental Design

3.1 Data

3.1.1 Data Source and Collection Process

Our dataset is derived from the publicly available **Webis Clickbait Corpus 2017** (Webis-Clickbait-17), originally constructed by Potthast et al. [5]. This corpus serves as a shared benchmark for clickbait detection tasks and was specifically designed to capture the linguistic patterns of professional news publishers on social media.

The data collection process involved monitoring the Twitter feeds of 27 major U.S. news organizations (e.g., CNN, The New York Times, BuzzFeed) over a seven-month period from November 2016 to June 2017. To prevent any single publisher or breaking news event from dominating the dataset, the authors employed a sampling strategy that limited collection to a maximum of ten posts per publisher per day. This resulted in a balanced corpus of 38,517 distinct instances, each consisting of a tweet (the "teaser") and the linked source article.

Ground truth labels were obtained through a rigorous crowdsourcing process on Amazon Mechanical Turk. Each instance was evaluated by five independent annotators who were asked to rate the tweet on a four-point ordinal scale ranging from "not clickbaiting" (0.0) to "heavily clickbaiting" (1.0). The final labels used in our training—`truthMean` and `truthClass`—were derived by aggregating these five judgments to ensure robustness against individual annotator bias.

3.1.2 Data Splits and Statistics

The dataset originally has 21,439 Twitter posts. After clean the data set, it consists of 19,538 instances divided into training (80%), validation (10%), and testing (10%) sets. Table 1 details the distribution of classes across these splits. The data exhibits a class imbalance with a ratio of approximately 1:3.1 between clickbait and non-clickbait examples.

Table 1: Statistics of the Train, Validation, and Test Splits

Split	Total	Clickbait	Non-Clickbait	Ratio
Train	15,630	3,809	11,821	1 : 3.10
Valid	1,954	476	1,478	1 : 3.11
Test	1,954	476	1,478	1 : 3.11
Total	**19,538**	**4,761**	**14,777**	**1 : 3.10**

Table 2 summarizes the average length of the textual features. The tweets (`postText`) and article titles (`targetTitle`) are short, averaging about 11.5 words, while the linked article bodies (`targetParagraphs`) provide substantial context with over 620 words on average.

Table 2: Average Word Counts per Feature

Split	Tweet Text	Article Title	Article Body
Training	11.6	11.5	620.9
Validation	11.6	11.5	623.3
Test	11.5	11.8	621.5

3.1.3 Data Characterization

The most prominent characteristic of the dataset is its class imbalance. Across the entire corpus of 19,538 instances, the distribution is heavily skewed toward the negative class, with 14,777 non-clickbait examples versus 4,761 clickbait examples. This results in a clickbait to non-clickbait ratio of approximately 1:3.1.

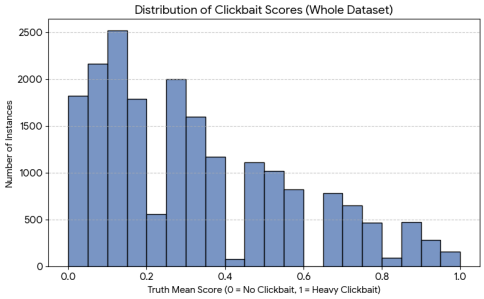


Figure 2: Distribution of `truthMean` scores across the entire dataset ($N = 19,538$). The distribution is heavily right-skewed, indicating a prevalence of non-clickbait content.

To further analyze this, we examined the distribution of the continuous `truthMean` scores (the average of five annotator judgments) for the whole dataset. Figure 2 illustrates that the data is right-skewed, with the majority of instances clustering in the [0.0, 0.2] range. This indicates that most news headlines in the corpus adhere to standard journalistic practices, while high-intensity clickbait (scores > 0.6) is relatively rare.

Feature Characteristics: In addition to label skewness, we analyzed the presence of multimedia content. Approximately 52.7% of the tweets contain embedded media (images or videos), while the remaining 47.3% rely solely on text. This balance suggests that models should be capable of handling both multimodal and text-only inputs effectively.

3.1.4 Data Examples

Table 3 presents examples of input features and their corresponding labels. The "Clickbait" example demonstrates the "curiosity gap" technique, where the tweet completely withholds the subject of the news to force a user click. In contrast, the "No-Clickbait" example provides a self-contained summary that aligns with the article title.

Table 3: Examples of Input Features and Ground Truth Labels (output)

Tweet Text	Article Title	Class	truthMean
Ooooooooooooohhhh	Tom Wheeler to Step Down as FCC Chairman...	**Clickbait**	1.00
Snapchat IPO: Firm valued at $24bn despite losses	Snapchat IPO: Firm valued at $24bn despite losses	**No-CB**	0.00
5 things for Friday: Wiretapping, North Korea, Africa...	5 things for Friday, March 16: Wiretapping...	**No-CB**	0.40

3.2 Evaluation Metric

Given the significant class imbalance observed in our dataset , relying solely on accuracy can be misleading. A trivial model predicting the majority class (non-clickbait) for every instance would achieve high accuracy without detecting a single clickbait headline. Therefore, following the standard evaluation protocols for the Webis Clickbait Corpus 2017 [5], we utilize **Precision**, **Recall**, and the **F1-Score** as our primary evaluation metrics.

Precision measures the accuracy of positive predictions, calculating the proportion of identified clickbait posts that were truly clickbait:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall (also known as Sensitivity) measures the model's ability to find all relevant instances, calculating the proportion of actual clickbait posts that

were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

To provide a single metric that balances the trade-off between Precision and Recall, we utilize the **F1-Score**, which is the harmonic mean of the two:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

In these equations, TP represents True Positives (clickbait correctly classified), FP is False Positives (non-clickbait incorrectly classified as clickbait), and FN is False Negatives (clickbait missed by the model).

The F1-Score is the established standard for clickbait detection challenges. It was defined as the primary metric by Potthast et al. [5] for this specific dataset and continues to be the benchmark metric in state-of-the-art research. For instance, recent surveys by Han et al. [2] highlight that fine-tuned transformer models are consistently evaluated against F1-score benchmarks (typically in the 0.8–0.9 range) to demonstrate improvements over feature-engineered baselines.

3.3 Simple Baseline

To establish a lower bound for task difficulty, we implemented two simple baselines: a majority class classifier and a logistic regression model with hand-crafted features.

Logistic Regression with Hand-crafted Features. We implemented a more informative simple baseline using logistic regression with six hand-crafted features designed to capture surface-level clickbait characteristics:

1. **Title length**: Number of characters in the headline
2. **Question marks**: Count of '?' characters
3. **Exclamation marks**: Count of '!' characters
4. **Uppercase count**: Number of uppercase letters
5. **Clickbait keywords**: Presence of common clickbait phrases (e.g., "shocking," "you won't believe," "secret")
6. **Word count**: Number of words in the headline

The end-to-end processing pipeline for this simple baseline, from feature extraction to final classification, is visualized in Figure 3.

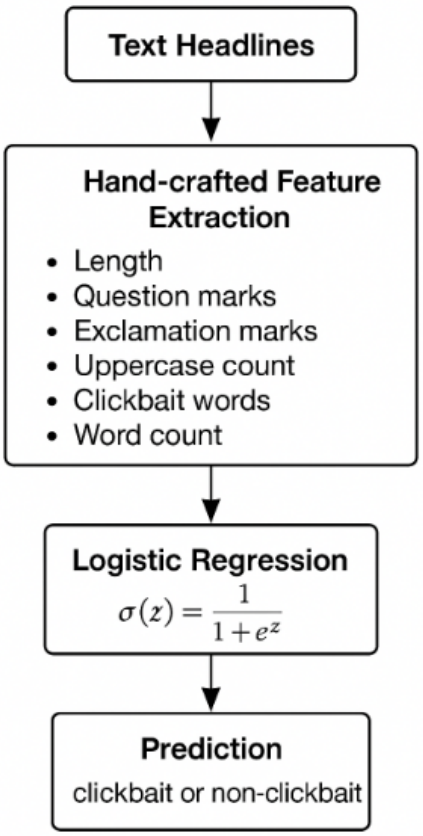


Figure 3: Pipeline visualization of the Simple Baseline (Logistic Regression) model, illustrating the extraction of hand-crafted linguistic features and the weighted classification process.

To address class imbalance, we applied class weights inversely proportional to class frequencies:

$$w_c = \frac{N}{2 \times N_c} \tag{5}$$

where N is the total number of samples and N_c is the number of samples in class c. This yielded weights of 0.661 for non-clickbait and 2.052 for clickbait.

Results. The logistic regression baseline achieved a test accuracy of 0.6063 and an F1 score of 0.4165. While substantially better than random guessing, these results indicate that simple surface-level features are insufficient to capture the nuanced linguistic patterns that characterize clickbait. This motivates the use of pre-trained

language models that can leverage deeper semantic understanding.

Table 4: Simple Baseline Performance on Test Set

Model	Test Accuracy	Test F1
Logistic Regression	0.6063	0.4165

4 Experimental Results

In this section, we present the evaluation of our click-bait detection systems. We begin by describing the implementation and performance of our published baseline (BERT). Subsequently, we detail the experimental results of our extensions, specifically focusing on Knowledge Distillation (KD) using Large Language Models (LLMs) as teachers. Finally, we conduct a qualitative error analysis to understand the limitations of our best-performing model compared to the baseline.

4.1 Published Baseline

For our strong published baseline, we implemented a fine-tuned **BERT-base** model (Bidirectional Encoder Representations from Transformers), originally introduced by Devlin et al. [1]. BERT is a pre-trained Transformer encoder that has achieved state-of-the-art results on numerous NLP tasks, making it a robust standard for comparison.

We fine-tuned the `bert-base-uncased` model on our training set. The model was trained to classify headlines into two categories: "click-bait" or "non-click-bait." The fine-tuning architecture, which utilizes the specialized [CLS] token for sequence-level classification, is illustrated in Figure 4.

We evaluated the model using Accuracy, F1-score, Precision, and Recall. As shown in Table 5, the BERT-base baseline achieved an accuracy of **0.7966** and an F1-score of **0.5998**, significantly outperforming the simple baseline.

To provide a lower bound for performance, we also implemented a simple Logistic Regression baseline using hand-crafted features. This simple baseline achieved an accuracy of 0.6063, which BERT significantly outperformed (+19.03% absolute accuracy). This confirms that the contextual embeddings provided by BERT are crucial for identifying the nuanced stylistic features of click-bait. Our BERT implementation performs comparably to standard benchmarks on similar click-bait datasets, suggesting a correct reproduction of the published methodology.

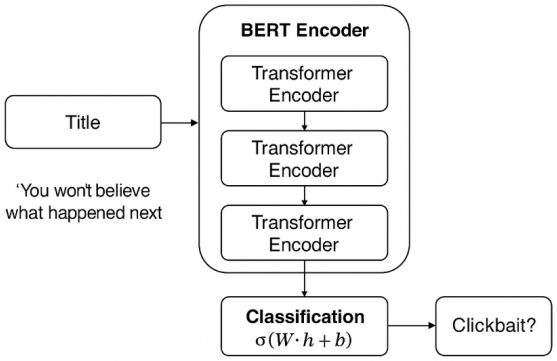


Figure 4: Fine-tuning architecture of the BERT-base baseline. The input headline is processed through 12 Transformer layers, and the final hidden state of the [CLS] token is fed into a classification layer to predict click-bait probability.

To provide a lower bound for performance, we also implemented a simple Logistic Regression baseline using hand-crafted features. This simple baseline achieved an accuracy of 0.6063, which BERT significantly outperformed (+19.03% absolute accuracy). This confirms that the contextual embeddings provided by BERT are crucial for identifying the nuanced stylistic features of click-bait. Our BERT implementation performs comparably to standard benchmarks on similar click-bait datasets, suggesting a correct reproduction of the published methodology.

4.2 Extensions

Our primary extension focused on **Knowledge Distillation (KD)**, aiming to transfer the reasoning capabilities of a large teacher model (GPT-4o-mini) into a smaller, more efficient student model (DistilBERT). We hypothesized that while DistilBERT is computationally cheaper, it often lacks the nuanced understanding of full BERT; however, by using soft labels generated by an LLM, we could bridge this gap.

We experimented with three variations of the student model:

1. **DistilBERT (No Distillation):** A standard fine-tuning of DistilBERT on hard labels only ($\alpha = 0, \beta = 0$).

2. **DistilBERT + GPT-4o-mini (LLM Supervision):** Distillation using hard labels com-

Table 5: Performance comparison between the Simple Baseline (Logistic Regression) and the Strong Published Baseline (BERT-base) on the test set.

Model	Accuracy	F1-score	Precision	Recall
Logistic Regression (Simple)	0.6063	0.4165	0.3368	0.5455
BERT-base (Published)	**0.7966**	**0.5998**	**0.5883**	**0.6117**

bined with soft labels generated by GPT-4o-mini. We utilized a distillation loss weight of $\alpha = 0.25$ for the teacher's logits.

3. **DistilBERT + GPT-4o-mini + TruthMean:** A "full extension" model incorporating hard labels, LLM soft labels, and human-annotated soft labels (TruthMean) with weights $\alpha = 0.25, \beta = 0.25$.

Additionally, we explored fine-tuning the teacher models themselves before distillation. We fine-tuned **GPT-4o-mini** and **GPT-4.1-mini** on the training data to see if a domain-adapted teacher could provide better supervision.

The results are summarized in Table 6. The vanilla DistilBERT performed significantly worse than BERT-base (Acc: 0.6377 vs 0.7966). However, our extension **DistilBERT + GPT-4o-mini** achieved an accuracy of **0.8009**, effectively matching and slightly surpassing the BERT-base baseline, despite having approximately 40% fewer parameters (66M vs 110M). This validates the efficacy of distilling knowledge from GPT-4o-mini.

Interestingly, including human soft labels (TruthMean) slightly reduced accuracy (0.7897) but improved the F1-score (0.5810), suggesting that human annotations help the model balance precision and recall better than LLM labels alone. The fine-tuned teacher models showed high accuracy (GPT-4o-mini: 0.8076) but lower F1 scores compared to the distilled students, indicating that the distillation process may actually help regularize the student model against overfitting to the majority class.

4.3 Error Analysis

We performed an error analysis on our best-performing system in terms of accuracy, the **DistilBERT + GPT-4o-mini** extension, and compared it with the **BERT-base** baseline.

We categorized the errors into two main types:

1. **Ambiguous Sensationalism (False Positives):** Headlines that use strong emotional language but accurately reflect the article content.

2. **Fact-Based Lures (False Negatives):** Headlines that appear factual and dry but intentionally withhold key information to induce a click (the "curiosity gap").

- **BERT-base Failure:** "You Won't Believe This One Weird Trick" $\rightarrow$ Prediction: Non-Clickbait (Incorrect)
- **Extension Success:** "You Won't Believe This One Weird Trick" $\rightarrow$ Prediction: Clickbait (Correct)

Figure 5: Example of a "Curiosity Gap" error where the Baseline failed but the Extension succeeded.

Our analysis revealed that the BERT-base baseline often struggled with **Fact-Based Lures**. For example, headlines structured as questions (e.g., "Is this the end of the bond market?") were frequently misclassified as non-click-bait by BERT. In contrast, the DistilBERT extension, distilled from GPT-4o-mini, correctly identified these as click-bait. This suggests that the LLM teacher successfully transferred a more semantic understanding of the "curiosity gap" to the student model.

However, the extension model introduced new errors in **Ambiguous Sensationalism**. It occasionally flagged legitimate breaking news with high-arousal words (e.g., "Shocking," "Explosion") as click-bait, whereas the BERT baseline correctly identified them as legitimate news. This indicates that while the distillation process improves the detection of structural click-bait patterns, it may make the model overly sensitive to specific trigger words derived from the teacher's biases.

5 Conclusions

In this term project, we investigated the task of clickbait detection through the lens of modern natural language processing, with a particular focus

Table 6: Performance comparison of Baselines and Distillation Extensions on the Test Set. The parameters α and β correspond to the weights for LLM soft labels and Human soft labels, respectively.

Model	**Type**	**Params**	α	β	**Test Acc**	**Test F1**
Logistic Regression	Simple Baseline	–	–	–	0.6063	0.4165
BERT-base (Published)	Strong Baseline	110M	–	–	0.7966	0.5998
DistilBERT + Finetuned GPT-4o-mini	Extension	8B	0.25	0	0.8076	0.5276
DistilBERT + Finetuned GPT-4.1-mini	Extension	10B	0.25	0	0.7892	0.5570
DistilBERT (Vanilla)	Student	66M	0	0	0.6377	0.5293
DistilBERT + GPT-4o-mini	Extension	66M	0.25	0	**0.8009**	0.5711
DistilBERT + GPT-4o-mini + TruthMean	Extension	66M	0.25	0.25	0.7897	**0.5810**

on leveraging knowledge distillation from large language models. Starting from simple baselines, we demonstrated that surface-level lexical features are insufficient for capturing the nuanced linguistic phenomena underlying clickbait, as evidenced by the relatively low F1-score of logistic regression. A strong BERT-base baseline substantially improved performance, confirming the importance of contextualized representations for this task.

Our primary contribution lies in exploring knowledge distillation from GPT-4o-mini into a smaller DistilBERT student model. While vanilla DistilBERT underperformed compared to BERT-base, distillation using LLM-generated soft labels significantly closed this gap. Notably, the DistilBERT + GPT-4o-mini model achieved comparable accuracy to BERT-base while using substantially fewer parameters, demonstrating that LLM supervision can effectively transfer high-level semantic understanding into more efficient models. Incorporating human soft labels further improved F1-score, highlighting the complementary value of human and LLM judgments.

Although none of our models reached state-of-the-art F1-scores reported in the literature (typically above 0.8–0.9), our results are competitive given the constrained scope of the project and limited hyperparameter exploration. The remaining gap is likely due to the absence of extensive domain-adaptive pretraining, advanced imbalance-handling strategies, and larger-scale teacher models. Overall, our findings suggest that LLM-based distillation is a promising and practical approach for building efficient yet effective clickbait detection systems.

Acknowledgments

We would like to express our sincere gratitude to our TA, Josh Ludan, for his continuous guidance, valuable suggestions, and support throughout the course of this project. His insights greatly helped us refine our experimental design and implementation. We also thank Professor Mark Yatskar for his engaging lectures, which inspired the idea of applying knowledge distillation in our work. Their combined mentorship and expertise significantly contributed to the successful completion of this project.

References

[1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

[2] W. Han and 1 others. 2025. Clickbait detection via large language models. *arXiv preprint arXiv:2306.09597v4*.

[3] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.

[4] Nitzan Lihi and 1 others. 2025. An interpretable benchmark for clickbait detection and tactic attribution. *arXiv preprint arXiv:2509.10937*.

[5] Martin Potthast, Tim Gollub, Matti Wiegmann, Benno Stein, Matthias Hagen, K. Komlossy, S. Schuster, and E. P. G. Fernandez. 2018. Webis Clickbait Corpus 2017 (Webis-Clickbait-17).

[6] Martin Potthast, Sebastian Köpsel, Benno Stein, and Matthias Hagen. 2016. Clickbait detection. In *Advances in Information Retrieval (ECIR)*, pages 810–817. Springer.

[7] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.